

Julian Juan

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EDUCATION

Los Gatos High School, Los Gatos, CA Class of 2027
4.0 GPA. AP Calculus BC, AP Physics C (Mechanics and E&M), AP Chemistry, AP Biology.

College coursework by dual and concurrent enrollment 2024 – present
West Valley College (Honors Program), Mission College, Laney College: 29 units, 4.0 GPA, Dean's List (Spring 2026).
Multivariable Calculus, Linear Algebra, Differential Equations, Statistics, AI. In progress: Machine Learning, Discrete Math.

RESEARCH EXPERIENCE

Research Intern, Institute of Physics, Academia Sinica, Taipei, Taiwan Summers 2025 – 2026

- Developed an ML surrogate for Geant4 calorimeter simulation; an early prototype ran about 200× faster in benchmarks.
- Built models that reconstruct a neutron's physical properties from thousands of calorimeter measurements.
- Analyzed nearly 1,800 oscilloscope captures to compare detector prototypes on reliability and timing (ROOT, C++, Python).

PROJECTS

Fast-MC-CBSC · generative surrogate for detector simulation · PyTorch

- Conditional generative model of neutron showers in a 6,790-channel calorimeter; 764,940 Geant4 events, 740 tests.
- Tested fidelity with a classifier two-sample test (AUROC 0.84 vs. a 0.65 target) and reported the gap as open.

PathoGraph-DL · spatio-temporal graph neural network · PyTorch Lightning, Vertex AI

- Forecasts crop-pathogen risk across ~194 countries: macro AUROC 0.960 ± 0.010 on a chronological held-out test.

Firefighter Tracker · indoor positioning where GPS fails · UWB ranging

- Multilateration built for noisy, blocked ranges, shaped by input from a Fremont Fire Department battalion chief.

LaTeXRenderer · collaborative LaTeX editor on a static host · TypeScript, Supabase

- Access decided server-side (Edge Functions, PBKDF2, signed URLs, SHA-256 asset checks); Playwright tests every bypass.

QTA0 · equity forecasting, reported as a negative result

- Re-tested an apparent trading edge under combinatorial purged cross-validation until it vanished (hit rate 52.5% to 49.7%).

HONORS

- **HiMCM** Finalist, team lead (one of ~7 U.S. finalist teams, 1,200+ worldwide); IMMC U.S. Candidate Round qualifier
- **Bay Area Data Science Competition**, Absolute Winner (Santa Clara University, Leavey School of Business)
- **Summer Science Program** invitee; **Conrad Challenge** Innovator; **President's Volunteer Service Award**, Gold
- **Santa Clara Valley Science & Engineering Fair**, First Place; California Science & Engineering Fair qualifier
- **Semifinalist**: Global Quantum Mechanics Challenge; Global Logic & Reasoning Competition (outstanding submission)

LEADERSHIP AND ACTIVITIES

- **Voting Student Member**, LGSUHS Budget Advisory Committee: student vote in oversight of an \$88M district budget; helped reconcile more than \$6M in cumulative reporting differences across four years
- **Founder and Co-President**, Game Theory Club, formed by merging the Chess Club and the Cards Club
- **Director of Outreach**, STEM Beyond Boundaries: taught about 80 children through interactive science lessons
- **Organizer**, Résumé Workshop, now part of the 9th–10th grade English curriculum; **Paid tutor** in math, science and history

SKILLS

Python, C++, TypeScript, PyTorch, scikit-learn, XGBoost, ROOT, Next.js, FastAPI, Supabase, Vertex AI, Zarr, Git, LaTeX